1. An one-dimensional simple harmonic oscillator with period T is found to be moving with velocity v_1 at $t = t$. After one quarter of the period, the oscillator is moving with velocity v_2. [5]
   (a) Find the amplitude of the oscillation. [3]
   (b) Plot v_1 versus v_2. [2]

2. A mass at the end of a spring oscillates with an amplitude of 5 cm at a frequency of 1 Hz. At $t = 0$ the mass is at its equilibrium position ($x = 0$). [5]
   (a) Find the position of the mass as a function of time. You may start from the generic solution without solving the equation of motion. [1]
   (b) What are the values of x, $\frac{dx}{dt}$, and $\frac{d^2x}{dt^2}$ at $t = \frac{1}{4}$ s? [3]
   (c) For any simple harmonic oscillator, show how the velocities at time t and time $t + T/2$ are related? [1]

3. A block of mass m is connected to a spring, the other end of which is fixed. There is also a viscous damping mechanism. The following observations have been made on this system: [5]
   1. If the block is pushed horizontally with a force equal to mg, the static compression of the spring is equal to h.
   2. The viscous resistive force is equal to mg if the block moves with a certain known speed u.

   (a) For this complete system, write the differential equation governing horizontal oscillations of the mass in terms of m, g, h, and u. [2]
   Answer the following for the case that $u = 3\sqrt{gh}$:
   (b) What is the angular frequency of the damped oscillation? [1]
   (c) After what time, expressed as a multiple of $\sqrt{h/g}$, is the energy down by a factor $1/e$? [1]
   (d) What is the Q factor of the oscillator? [1]

4. Consider an under-damped simple harmonic oscillator (of mass m, natural angular frequency ω_o, damping coefficient $\gamma = \omega_o/2$) subjected to a force $F_o \cos \omega t$. [5]
   (a) Describe the motion when $\omega = 0$, and the oscillator starts from rest with some non-zero amplitude. You do not need to solve any differential equation. [2]
   (b) Show that the steady-state amplitude (after the transients decayed to zero) is larger for $\omega = \omega_o$ than $\omega = 2\omega_o$. [2]
   (c) What is the Q factor of this oscillator? [1]